GROUP ASSIGNMENT 2

Automatic Evaluation and Error Analysis of an English–Indonesian Machine Translation System

BLEU-1 to BLEU-4, chrF and TER on FLORES-200, comparing MarianMT and NLLB

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Chapter 13: Machine Translation

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1. Introduction

This report evaluates two pre-trained neural machine translation (NMT) systems on English-to-Indonesian translation. The aim is twofold. First, to measure translation quality with the standard automatic metrics (BLEU-1 to BLEU-4, chrF and TER) and to see how the length of the n-gram changes BLEU. Second, to look at the sentences the metrics score worst and ask what actually went wrong, using what is known about the differences between English and Indonesian.

The main finding is that the two systems differ by only a few points, and that the metrics are better at ranking systems than at telling us whether an individual translation is good. Several of the lowest-scoring sentences are not the worst translations, and several of the worst translations are not far below the average score. Sections 5 and 6 go through the evidence.

2. Data, Models and Metrics

2.1 Data

The test data come from the FLORES-200 English–Indonesian parallel set supplied for the assignment (2,009 sentence pairs, one human reference per sentence). Some references in the file had broken character encoding (for example *BarÃ$a* for *Barça* and *SÃ£o Paulo* for *São Paulo*, 50 lines in total). These were repaired before scoring so that correct output is not penalised for accents. A random sample of **200 pairs** was drawn with a fixed seed (42), so the experiment can be repeated on exactly the same sentences.

2.2 Systems

Two pre-trained models from HuggingFace were used without any fine-tuning, both decoding with beam search (4 beams):

- **MarianMT** (Helsinki-NLP/opus-mt-en-id), a compact Transformer trained on OPUS data.
- **NLLB-200 distilled 600M** (facebook/nllb-200-distilled-600M), a larger multilingual Transformer covering 200 languages, with the language codes eng_Latn to ind_Latn.

Both are Transformer encoder–decoder models. The encoder turns the English sentence into a sequence of contextual vectors using self-attention. The decoder then produces the Indonesian sentence one token at a time, and at each step it attends to the encoder output through cross-attention, which is how it decides which English words to translate next. Translation was run on a Google Colab T4 GPU.

2.3 Metrics

- **BLEU-n** (cumulative): the geometric mean of the modified 1-gram to n-gram precisions, multiplied by a brevity penalty. BLEU-1 only checks that the right words appear; BLEU-4 also needs four consecutive words to match.
- **chrF**: an F-score over character n-grams up to length 6, weighting recall twice as much as precision. **chrF++** adds word unigrams and bigrams. Because it works on characters, it gives partial credit to words that share a stem but differ in affixes, which matters for Indonesian.

- **TER**: the number of edits (insertions, deletions, substitutions and block shifts) needed to turn the output into the reference, divided by the reference length. Lower is better, and it is the only metric here where that is the case.

All scores were computed with sacreBLEU (13a tokenisation) at corpus level. For the error analysis, sentence-level BLEU (smoothed), chrF and TER were also computed. Sentence-level BLEU is noisy because many sentences have no matching 4-gram at all, so it was used only to find candidate sentences, not to compare systems. 95% confidence intervals come from bootstrap resampling of the sentences (300 resamples).

3. Results

Metric	MarianMT	NLLB-600M
BLEU-1	65.87	**69.22**
BLEU-2	53.17	**56.50**
BLEU-3	43.72	**46.81**
BLEU-4	36.43	**39.33**
chrF	63.46	**66.13**
chrF++	61.14	**63.88**
TER (lower is better)	48.55	**44.18**

Table 1. Corpus-level scores on 200 FLORES-200 sentences. Best value per row in bold.

NLLB is ahead of MarianMT on every metric: about 3 points on BLEU-1 to BLEU-4, 2.7 points on chrF and 4.4 points lower on TER. Table 2 gives the bootstrap intervals for each system.

System	Metric	Score	CI low	CI high
MarianMT	BLEU-1	65.87	63.69	67.84
MarianMT	BLEU-4	36.43	33.63	39.15
MarianMT	chrF	63.46	61.44	65.13
MarianMT	TER	48.55	45.96	50.99
NLLB-600M	BLEU-1	69.22	67.05	71.07
NLLB-600M	BLEU-4	39.33	36.26	42.51
NLLB-600M	chrF	66.13	64.40	68.08
NLLB-600M	TER	44.18	41.53	46.79

Table 2. 95% bootstrap confidence intervals.

Each system's own interval is fairly wide, and the two overlap (for BLEU-4, 33.6–39.2 against 36.3–42.5). Overlapping separate intervals do not rule out a real difference, because both systems were scored on the same 200 sentences. A paired bootstrap that resamples sentences once and scores both systems on the same resample is the better test. It gives a BLEU-4 advantage for NLLB of **2.90 points (95% CI 0.94 to 4.66)** and a chrF advantage of **2.67 points (95% CI 1.39 to 3.89)**, and a TER that is **4.37 points lower (95% CI −6.37 to −2.49)**. None of the intervals includes zero, so on this sample NLLB is better than MarianMT on all three metrics, although the size of the gaps is uncertain (for BLEU-4, roughly 1 to 5 points).

4. Effect of n-gram Length on BLEU

	n = 1	n = 2	n = 3	n = 4
MarianMT BLEU-n	65.87	53.17	43.72	36.43
NLLB BLEU-n	69.22	56.50	46.81	39.33
MarianMT precision (%)	69.0	44.9	31.0	22.1
NLLB precision (%)	71.9	47.9	33.4	24.2

Table 3. Cumulative BLEU-n and individual n-gram precision. Brevity penalty: 0.955 (MarianMT), 0.962 (NLLB).

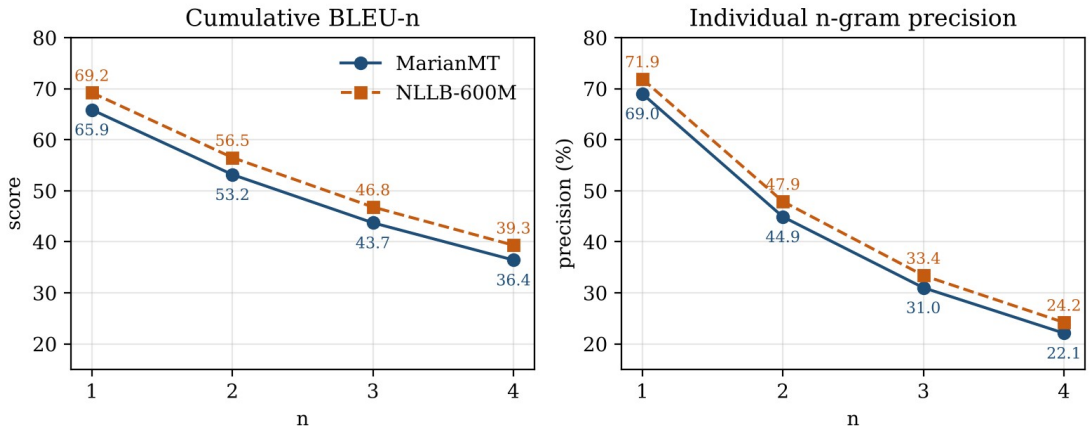


Figure 1. BLEU falls steadily as the n-gram order increases, for both systems.

BLEU drops sharply with n. For MarianMT it goes from 65.9 at n = 1 to 36.4 at n = 4, a loss of 29.4 points or about 45% of the starting value. NLLB loses a similar amount, 29.9 points or 43%, so the shape is almost identical and NLLB is simply shifted up. Looking at the individual precisions shows where the loss comes from. About 69–72% of the output words appear in the reference, but only 22–24% of the four-word sequences do. From one order to the next, the precision is multiplied by roughly 0.65–0.72, which means each extra word of required context removes around a third of the matches that were left.

This is expected, and part of the explanation is linguistic. A unigram match only asks whether the translator picked an acceptable word. Longer matches also require the same word order, the same affixes and the same function words (*yang*, *di*, *dari*, *untuk*) in the same positions, and Indonesian leaves a lot of freedom there. A sentence can be translated with an active or passive verb, with or without *yang*, with *para* or reduplication for plurals, and each choice breaks every n-gram that crosses it. The brevity penalty of 0.955 to 0.962 shows that the output is also about 4% shorter than the references overall, which is a small cost compared with the loss from higher orders and is mostly explained by the omissions discussed in Section 5.

The practical consequence is that BLEU-1 is close to a vocabulary score, while BLEU-4 is the strictest and the one most affected by legitimate variation. Reporting the whole range, as done here, is more informative than a single BLEU-4 figure.

5. Qualitative Error Analysis

The 10 lowest-scoring sentences (with at least six words) were taken for MarianMT, ranked by the combined rank of sentence-level BLEU and chrF. All 10 are analysed below. Each was read against its reference and classified by linguistic phenomenon. Sections 5.3 and 5.4 add further examples from both systems and the ten lowest-scoring NLLB sentences. Sentence numbers are the identifiers in the FLORES file.

5.1 The ten lowest-scoring sentences (MarianMT)

Sentence 375 sBLEU 1.5, chrF 18.7, TER 100.0

EN	A person who weighs 200 pounds (90kg) on Earth would weigh about 36 pounds (16kg) on Io. So the gravity, of course, pulls less on you.
Ref	Orang yang berbobot 90 kg di Bumi akan berbobot 16 kg di Io. Itu sebabnya gravitasi memiliki daya tarik lebih rendah.
MT	Jadi gravitasi, tentu saja, mengurangi berat badan Anda.

Phenomenon: *Omission of a whole sentence; meaning shift*

The system translated only the second English sentence and dropped the first one completely, so the output contains none of the numbers. Even the sentence it did translate changes meaning: *pulls less on you* became *mengurangi berat badan Anda* (“reduces your weight”), a cause where the source states an effect. The reference is also not literal. It keeps only the metric values and drops the pound figures, which explains part of the very low score. TER is 100 because nearly the whole reference has to be rewritten.

Sentence 355 sBLEU 5.3, chrF 30.6, TER 76.0

EN	The blade on a modern ice skate has a double edge with a concave hollow between them. The two edges allow for a better grasp of the ice, even when tilted.
Ref	Bilah pada seluncur es modern memiliki sisi ganda dengan cekungan kosong diantaranya. Dua sisi ini memungkinkan cengkeraman yang lebih baik terhadap es, bahkan ketika dimiringkan.
MT	Pisau pada skate es modern memiliki tepi ganda dengan cekung berongga antara mereka.

Phenomenon: *Polysemy, lexical gap, pronoun mismatch, omission*

Three separate problems. *Blade* is rendered as *pisau* (a kitchen or pocket knife) instead of *bilah*, the sense needed for a skate: a polysemy error. *Ice skate* is left as the loan word *skate es*, while the reference uses the established Indonesian term *seluncur es*, so the system fell back on borrowing where a native term exists. *Between them* becomes *antara mereka*. In Indonesian, *mereka* normally refers to people, so using it for two edges sounds like a literal English calque, whereas the reference uses the clitic form *diantaranya*. The second sentence is dropped entirely.

Sentence 555 sBLEU 6.0, chrF 37.8, TER 83.3

EN	Products can be purchased as needed, but most will have little or no real impact on performance.
Ref	Produk-produk dapat dibeli sesuai dengan kebutuhannya, tetapi sebagian besar akan sedikit atau sama sekali tidak berdampak pada performa.
MT	Produk dapat dibeli seperti yang diperlukan, tetapi kebanyakan tidak memiliki dampak yang nyata terhadap kinerja.

Phenomenon: *Acceptable alternative wording, with one small loss of meaning*

This is mostly a good translation that BLEU punishes. *Kebanyakan* for *sebagian besar*, *kinerja* for *performa* and the bare *produk* for the reduplicated *produk-produk* are all acceptable choices, and *seperti yang diperlukan* is only a more literal way of saying *as needed* than the reference's *sesuai dengan kebutuhannya*. There is one real loss: *little or no impact* became *tidak memiliki dampak yang nyata* ("no real impact"), so the "little" half of the contrast is gone. Most of the low score comes from synonym choice and phrasing, not from error.

Sentence 1396 sBLEU 6.9, chrF 39.9, TER 73.1

EN	Greenland was settled sparsely. In the Norse sagas they say that Erik the Red was exiled from Iceland for murder, and when travelling further west, found Greenland and named it Greenland.
Ref	Greenland berpenduduk jarang. Di saga Nordik dikatakan bahwa Erik si Merah diasingkan dari Islandia akibat pembunuhan, dan saat pergi ke barat menemukan Greenland dan menamakannya Greenland.
MT	Greenland diselesaikan secara perlahan-lahan. dan ketika bepergian ke barat, menemukan Greenland dan menamainya Greenland.

Phenomenon: *Polysemy (settle, sparsely) and omission*

Settled was read as "completed" (*diselesaikan*) instead of "inhabited", and *sparsely* was read as "slowly" (*perlahan-lahan*) instead of "thinly". Both are word-sense errors, and together they turn a statement about population into a nonsensical one about a country being finished slowly. The sentence about Erik the Red is dropped, leaving a fragment that starts with a lowercase *dan* after a full stop. The reference solves *was settled sparsely* with a different structure (*berpenduduk jarang*, "has a sparse population"), which no word-by-word match could produce.

Sentence 1041 sBLEU 7.0, chrF 40.4, TER 100.0

EN	The speeds of 802.11n are substantially faster than that of its predecessors with a maximum theoretical throughput of 600Mbit/s.
Ref	802.11n jauh lebih cepat daripada generasi sebelumnya dengan throughput teoretis maksimum 600Mbit/detik.
MT	Kecepatan 802.11n secara substansial lebih cepat dari pendahulunya dengan sejumlah besar sejumlah besar dari 600Mbit/s.

Phenomenon: *Word order (M-D vs D-M), technical term, repetition error*

The noun phrase *maximum theoretical throughput* is a stack of modifiers before the head. Indonesian puts the head first, so the reference has *throughput teoretis maksimum*. The system did not manage this restructuring and instead produced *sejumlah besar sejumlah besar* ("a large number, a large number"), a repeated phrase that has no source in the English and loses the technical meaning altogether. *Substantially faster* was translated word for word as *secara substansial lebih cepat*, which is stiff, while the reference simply says *jauh lebih cepat*. The unit *Mbit/s* was kept, while the reference localises it to *Mbit/detik*.

Sentence 655 sBLEU 7.4, chrF 41.1, TER 68.2

EN	Crime, and ill-will toward foreigners in general, is virtually unknown in Greenland. Even in the towns, there are no "rough areas."
Ref	"Kejahatan dan rasa dengki terhadap orang asing pada umumnya, sebenarnya tidak dikenal di Greenland. Bahkan, di dalam kota tidak ada "daerah kasar"."

MT	Kejahatan, dan sakit terhadap orang asing secara umum, hampir tidak diketahui di Greenland.

Phenomenon: *Polysemy (ill-will) and omission; problem in the reference*

Ill-will is translated as *sakit* (“sick, in pain”), taking *ill* in its health sense. The right meaning is a feeling of hostility, which the reference renders as *rasa dengki*. The second sentence is again missing. It is also worth noting that the reference is not clean: it opens and closes with stray quotation marks that the English does not have. A small part of the penalty therefore comes from the reference, not the system.

Sentence 1107 sBLEU 4.5, chrF 46.3, TER 64.3

EN	She directed officers to her black Oldsmobile Intrigue which was 500 feet away.
Ref	Ia mengarahkan para petugas ke Oldsmobile Intrigue berwarna hitam miliknya yang berjarak 152 meter.
MT	Dia mengarahkan petugas untuk Oldsmobile hitam nya Intrigue yang 500 meter jauhnya.

Phenomenon: *Number/unit error (serious), word order and named entity*

This is the most serious error among the ten, even though it costs only one token. *500 feet* became *500 meter*, which changes the distance by a factor of about 3.3. The reference converts the unit correctly to *152 meter*. The colour adjective was placed after the noun, as Indonesian requires, but in the wrong place: it splits the model name, giving *Oldsmobile hitam nya Intrigue* instead of keeping *Oldsmobile Intrigue* together, and the possessive *-nya* is written as a separate word. The preposition is also wrong: *mengarahkan … untuk* (“for”) where the reference has *ke* (“to”). Because the automatic flag for this sentence was empty, it is a good reminder that the heuristics miss things that reading the sentence finds at once.

Sentence 1377 sBLEU 8.0, chrF 42.1, TER 61.5

EN	Plants make their food from the sun by photosynthesis. They also provide shade.
Ref	Tumbuhan membuat makanan mereka dari matahari melalui fotosintesis. Mereka juga memberikan tempat berteduh.
MT	Tanaman membuat makanan dari matahari dengan fotosintesis.

Phenomenon: *Omission; near-synonym choice*

The second sentence (*They also provide shade*) is dropped, which alone removes about a third of the reference. In the part that was translated, *plants* became *tanaman* instead of *tumbuhan*. Both can mean “plant”, but *tanaman* usually refers to cultivated plants, while *tumbuhan* is the general botanical term, so the reference choice fits the science context better. The possessive *their* (*mereka*) is also left out, which is acceptable in Indonesian.

Sentence 1495 sBLEU 6.5, chrF 45.5, TER 76.7

EN	Subsistence agriculture is a simple, often organic, system using saved seed native to the ecoregion combined with crop rotation or other relatively simple techniques to maximize yield.
Ref	Pertanian subsisten merupakan sistem yang sederhana, biasanya organik, dan memakai benih tersimpan asal ekoregion tersebut, yang digabung dengan penanaman bergilir maupun teknik lainnya yang relatif simpel guna memaksimalkan hasil panen.
MT	Pertanian penggantian adalah sistem yang sederhana dan seringkali organik menggunakan benih asli yang disimpan untuk lingkungan dikombinasikan dengan rotasi tanaman atau teknik sederhana lainnya untuk memaksimalkan hasil.

Phenomenon: *Domain terminology and structural disparity*

Subsistence is translated as *penggantian* (“replacement”), so the domain term *pertanian subsisten* is lost. *Ecoregion* is generalised to *lingkungan* (“environment”), and *native to* becomes *untuk* (“for”), which changes the meaning. Structurally, the English sentence strings together reduced clauses (*using … combined with …*). Indonesian generally marks such links with *dan* or *yang*, and the reference does exactly this (*dan memakai … yang digabung dengan …*). The system keeps the English chain of participles (*menggunakan … dikombinasikan …*), which is grammatical but reads like a calque and shares few n-grams with the reference. This is the structural disparity the assignment mentions, in a long sentence.

Sentence 487 sBLEU 6.5, chrF 47.1, TER 88.2

EN	Sikhs consider their faith to be a separate religion from Hinduism though they acknowledge its Hindu roots and traditions.
Ref	Sikh menganggap kepercayaan mereka berbeda dari Hinduisme, walau mereka juga mengakui akar dan tradisi mereka dalam Hindu.
MT	Orang Sikh menganggap iman mereka sebagai agama yang terpisah dari agama Hindu meskipun mereka mengakui asal usul dan tradisi Hindunya.

Phenomenon: *Valid paraphrase: lexical variation, not an error*

The translation is accurate and fluent. *Orang Sikh*, *meskipun* for *walau*, *asal usul* for *akar* and the possessive *Hindunya* are normal Indonesian, and *agama Hindu* is arguably more usual than *Hinduisme*. The one choice that deserves a comment is *iman* for *faith*, since *iman* leans toward religious belief in an Islamic sense while *kepercayaan* is neutral. Even so, the sentence is acceptable, and it scores TER 88.2 and sBLEU 6.5. This is the clearest example in the sample of BLEU measuring similarity to one translator’s wording rather than quality.

5.2 Patterns across the ten sentences

Phenomenon	Count	Sentences
Omission of a whole sentence	5	375, 355, 1396, 655, 1377
Polysemy / wrong word sense	4	355 (blade), 1396 (settled, sparsely), 655 (ill-will), 1495 (subsistence)
Structural disparity / word order (M-D vs D-M)	3	1041, 1107, 1495
Lexical gap / technical term	3	355 (ice skate), 1041 (throughput), 1495 (ecoregion)
Number or unit error	2	375, 1107
Acceptable alternative wording (not an error)	2	555, 487

Table 4. Phenomena found in the ten lowest-scoring MarianMT sentences (a sentence can have more than one).

The most striking pattern is omission. In five of the ten sentences the input contained two or three sentences, and MarianMT translated only some of them. This is not limited to the worst ten. Only 29 of the 200 test inputs contain more than one sentence, but MarianMT’s output is shorter than 75% of the reference length for 12 of those 29 (41%), against 5 of the remaining 171 single-sentence inputs (3%). Their average length ratio (output words divided by reference words) is 0.80 for the multi-sentence inputs and 1.01 for the single-sentence ones. NLLB shows the same weakness more mildly: 5 of 29 multi-sentence inputs are shorter than 75% of the reference, and the average ratio is 0.91. A

likely reason is that these models were trained mostly on single sentences. It matters for evaluation because splitting the input into sentences before translating would probably remove a large part of the low scores, and this is a processing issue and not a limitation of the Indonesian output itself.

Among the genuine linguistic errors, word sense is the most common. Words such as *blade*, *settle*, *ill-will* and *subsistence* each have several meanings or a narrow technical one, and the system picked a frequent sense that does not fit the context. Word order errors appear where English stacks modifiers in front of a noun (*maximum theoretical throughput*, *her black Oldsmobile Intrigue*). Indonesian needs the head first and a looser structure for the modifiers, and the output tends to keep the English shape. Lexical gaps show up with technical terms, where the reference uses a loan word or an established coinage and the system chooses either a wrong literal word or a different loan. Finally, the unit error in sentence 1107 shows the kind of mistake that carries real consequences while costing the same one token as a harmless synonym.

5.3 Further examples, including NLLB

The table below adds sentences from the comparison of low BLEU and high chrF, covering both systems. They show phenomena that the ten sentences above did not.

ID	System	Observation	Phenomenon
352	Both	Reference *tegangan permukaan*. Marian: *ketegangan permukaan*; NLLB the same. Marian *mengapung*, NLLB *melayang* for “float”.	Polysemy: *tension* in the physics sense versus *ketegangan* (stress). *Melayang* suggests floating in the air, not on water.
136	Both	Reference *bersinggungan*. Marian: *berpotongan*; NLLB: *bersilang*. Both write *alami* or *dramatis* where the reference has *alam* and *drastis*.	Lexical choice for *intersect*; near-synonyms and affix variation
766	Marian	Output has *anak - anak* and *kamar - kamar* with spaces around the hyphen; chrF 85.2 but sBLEU 21.5. *Hostel* became *asrama*.	Reduplication spacing (a formatting artefact); lexical gap (*hostel* vs *asrama*, a dormitory)
814	NLLB	Reference *berjalan kaki seharian* (“a whole day”). NLLB: *sehari-hari* (“everyday”). It ends with *menikmati mereka* where the reference has *menikmatinya*.	Wrong sense (*day-long*); pronoun: *mereka* (people) used for things instead of *-nya*
1450	NLLB	sBLEU 4.6 but chrF 56.8. *Periode-periode ini* (reduplicated plural) and *antara banyak bangsawan* (“between many nobles”) against the reference’s *para bangsawan*.	Plurality: reduplication or *banyak* versus the plural marker *para*; reduplication where a single form is enough
1266	NLLB	sBLEU 32.9 but chrF 87.1. Differences are *untuk*, *tanggal* and *Magistrates* for *Magistrate*.	Insertion of function words and a proper-noun variant; nearly correct

Table 5. Additional sentences from both systems.

Two points come out of these. First, the physics term in sentence 352 (*surface tension*) is mistranslated identically by both systems, as *ketegangan*, which suggests a training-data gap for this term and not a quirk of one model. Second, the pronoun error in sentence 814 comes from a difference in the two languages: English *them* can refer to things, while Indonesian *mereka* is normally reserved for people, and inanimate references need *-nya* or a repeated noun.

5.4 The ten lowest-scoring NLLB sentences

The same selection was made for NLLB (Table 6). Sentences 355 and 1495 are also on the MarianMT list and were discussed in Section 5.1; NLLB makes the same *pisau* (blade) and *skate* errors in 355, and in 1495 it writes *susistensi* (a misspelling of *subsistensi*) where the reference has *subsisten*.

ID	What happens (reference vs NLLB)	Phenomenon
355	Reference *Bilah … seluncur es … cekungan kosong diantaranya*; NLLB *Pisau … skate es … lubang konkaf di antara keduanya*. The second sentence is dropped.	Polysemy (*blade*), lexical gap (*ice skate*), omission. Using *keduanya* for "the two of them" is better than Marian's *mereka*.
1042	Source: *Duvall, who is married with two adult children, did not leave a big impression on Miller, to whom the story was related.* NLLB says *Miller, yang menceritakan kisahnya* ("Miller, who told the story").	Wrong relative-clause attachment: *to whom the story was related* means the story was told to Miller, but the output makes Miller the teller (agent and recipient reversed). The first clause is translated well and differs from the reference only in wording.
533	NLLB *mengambil sedikit strokes, atau berayun dari klub … menang*; reference *melakukan pukulan atau ayunan stik paling sedikit … Pemenangnya adalah pemain.*	Lexical gap and polysemy: *strokes* left in English, *club* rendered as *klub*, which usually means an organisation, instead of *stik* (golf club), *course* as *kursus* (a study course). *Fewest* became *sedikit* ("a few"), losing the superlative.
1340	Source: *It can be pulled into thin wire, which can be twisted and plaited. It can be hammered or rolled into sheets.* NLLB translates only the first sentence and renders *plaited* as *dilapisi* ("coated"). The reference supplies the subject *Emas* (gold) from the previous sentence.	Omission of the second sentence; wrong word sense (*plait* vs *coat*); reference adds context that the source does not contain.
1199	NLLB leaves *men standing Super-G* and *men sitting Super-G* untranslated and writes *Czech* instead of *Ceko*; reference *Super-G berdiri pria*.	Untranslated English (code-switching) in a term made of modifiers; word order again: English *men's standing Super-G* needs the head first, *Super-G berdiri pria*. *Pemain Australia* for *Australia's* also adds a word that is not in the source.
1570	NLLB translates the first sentence only. It also writes *25 sisi lebar Dunlap* for *the 25 Dunlap broadsides* (a literal reading of *broadside* as "wide side"); the reference has *selebaran*.	Omission; polysemy / lexical gap (*broadside* as a printed sheet).
1152	Source: *Looking decidedly rusty … South Africa however steadily improved.* NLLB: *Afrika Selatan terlihat sangat berkarat* ("looked very covered in rust"); reference *kurang latihan* ("out of practice").	Idiom translated literally (*rusty*). *Saudara-saudaranya di selatan* ("their brothers in the south") for *southern sisters* (a team nickname) is another literal reading. The sentence order is also changed.
1808	NLLB *pemalu* ("shy") for *shy*, *mereka* for the animals, and *binocular* for *binoculars*; reference *tidak ingin menampakkan diri, dapat dilihat, teropong*.	Literal rendering of a figurative use of *shy*; *mereka* used for animals (see Section 5.3); loan word where a native term (*teropong*) exists.
675	NLLB *Masuk ke Afrika Selatan* for *Entering Southern Africa*; reference *wilayah selatan Afrika*.	Proper-name ambiguity: *Southern Africa* is a region, not the country *Afrika Selatan* (South Africa), a factual change. Other differences are valid wording: *luar biasa* vs *menakjubkan*, *rute wisata normal* vs *rute turis*.

Table 6. The ten lowest-scoring NLLB sentences (ranked by sentence BLEU and chrF).

NLLB's errors differ from Marian's in one respect. It rarely produces nonsense, and its low scores come more often from reading an English expression literally (*rusty*, *shy*, *plaited*, *broadside*), leaving

specialised terms untranslated, or choosing the wrong referent (sentences 1042 and 675). It still drops whole sentences in multi-sentence inputs (355, 1340, 1570). The picture across both systems is that vocabulary with several meanings, English noun-phrase structure and sentence boundaries are the weakest points.

6. Limits of BLEU Compared with Actual Translation Quality

The corpus scores in Section 3 are useful for comparing systems, but the sentences in Section 5 show where they stop being reliable. The sentence-level BLEU and chrF correlate strongly (Pearson r = 0.85 for MarianMT and 0.86 for NLLB), so the two metrics largely agree on which sentences are weak. What the examples show is where all of them can mislead.

- **Synonyms and paraphrase get no credit.** Sentence 487 is accurate and natural but scores sBLEU 6.5 because almost every content word differs from the single reference. Sentence 555 behaves the same way.
- **One reference is a narrow target.** FLORES gives one translation per sentence, and Indonesian offers several acceptable ones: *Hinduisme* or *agama Hindu*, *kinerja* or *performa*, *tumbuhan* or *tanaman*, active or passive voice, with or without *yang*. BLEU treats every departure from the reference as an error, and it measures closeness to that one translator's wording instead of adequacy.
- **The reference is not always literal or clean.** In sentences 375 and 1107 the reference converts or drops units, and in 655 it has stray quotation marks. A system that translates faithfully is penalised for not matching those choices.
- **Morphology is handled coarsely.** Indonesian relies on affixes (*me-, di-, ber-, -kan, -nya*), so *mengumumkan* against *diumumkan* or *Hindu* against *Hindunya* are complete misses for BLEU while chrF gives partial credit. This is why chrF scores sentence 766 at 85.2 and BLEU at 21.5.
- **Error severity is invisible.** The wrong unit in sentence 1107 (*500 meter* for 500 feet) and the wrong word *sakit* in 655 each cost about as little as a harmless synonym. A dropped negation or a wrong number can reverse the meaning of a sentence and still leave the score nearly unchanged.
- **Fluency and global structure are barely checked.** Only sequences of up to four words are matched. A broken sentence such as the repeated *sejumlah besar* in 1041 can still receive some credit from the correct words around it.
- **Formatting artefacts distort scores.** The spaces in *anak - anak* (sentence 766) are not a translation error, yet they cut BLEU from what would be a much higher value.
- **Sentence-level BLEU is unstable.** Many sentences have no matching 4-gram, so the score is dominated by smoothing. It was useful here only to pick candidates, and the final judgement needed reading the text.

For these reasons, chrF and TER are useful companions to BLEU. chrF is more tolerant of affix variation, and TER can be read as a rough estimate of post-editing effort. Neural metrics such as BERTScore or COMET, and above all human judgement of adequacy and fluency, are still needed to say whether a translation is actually good. In this sample, the ranking from BLEU (NLLB first) agrees with the ranking from chrF and TER, so system-level conclusions look stable, while the per-sentence picture is much less so.

7. Conclusion

On 200 randomly sampled FLORES-200 sentence pairs, NLLB-200 (600M) scored higher than MarianMT on every metric: BLEU-4 39.3 against 36.4, chrF 66.1 against 63.5, and TER 44.2 against 48.6. A paired bootstrap confirms the advantage for BLEU-4 (+2.9, 95% CI 0.9 to 4.7), chrF (+2.7, 95% CI 1.4 to 3.9) and TER (−4.4, 95% CI −6.4 to −2.5), though the gaps are modest.

BLEU fell by about 29–30 points, roughly 45% of its value, between BLEU-1 and BLEU-4 for both systems. Individual precision fell from about 70% for single words to about 23% for four-word sequences, which reflects how much order, function words and affixation can vary in valid Indonesian translations.

The lowest-scoring MarianMT sentences were caused mainly by omitted sentences in multi-sentence inputs, wrong word senses (*blade*, *settle*, *ill-will*, *subsistence*), word order problems around stacked English modifiers, technical terms without a good equivalent, and number or unit errors. Two of the ten were acceptable translations that simply differed from the reference. Automatic metrics are therefore a reasonable guide for comparing systems on a whole test set, but they cannot replace a linguistic reading of the output, particularly for judging how serious an error is.

Use of AI Assistance

AI assistance was used in preparing this project, including for its structure and layout.

Appendix: Reproducibility

- Code, data and results: https://github.com/Mutaib-yye/translate_machine (notebook notebooks/MT_Evaluation_EN_ID.ipynb).
- Settings: 200 sentences, seed 42, beam size 4, batch size 16, sacreBLEU with 13a tokenisation, 300 bootstrap resamples, BERTScore not used.
- Models: Helsinki-NLP/opus-mt-en-id and facebook/nllb-200-distilled-600M; run on a Google Colab T4 GPU.